

A New Self-Adaptive Bionic Ankle with 2 DoF

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Abstract

To build humanoid robots that have the ability to transverse through complex terrains, I designed a self-adaptive ankle with 2 DoF. It uses two motors as the power to drive the leg and universal joint to act as the real joint of a human ankle, it can move in a plane, being able to achieve dorsiflexion up to 24.1° , plantar flexion up to 33.7° , inversion up to 21.9° , and eversion up to 14.6° , and also a spring system that can absorb shock. Using additional distance sensors, it have the ability to calculate the pitch and roll of the foot relative to the ground, able it move the foot to become parallel to the ground using PID controller algorithm. This makes it able to balance itself on uneven terrain, and land perfectly onto areas with inclination swiftly.

Contents

1 Introduction	3
1.1 Background and Significance	3
1.2 Predecessors	3
1.2.1 An Energy Storage and Return Bionic Feet	3
1.2.1 Hugh Herr's bionic ankle	4
1.2.2 A Multi-loop Ankle Foot	4
1.3 Improvements	5
2 Theoretical Design	6
2.1 Correspondence to a real human leg	6
2.2 Spring System	7
2.3 3D-printed Legs	7
2.4 Universal Joint	8
2.5 Drive System	8
2.6 3D-printed Foot	8
2.7 Conclusion	8
3 Structural Design	10
3.1 Introduction	10
3.2 Drive Leg	10
3.3 Connector Leg	12
3.4 Foot	13
3.5 Wire rope	13
3.6 Conclusion	14
4 Hardware Design	16
4.1 Introduction	16
4.2 JY62	17
4.7 VL53L0X	17
4.5 Esp32	17
4.2 Motor MS9015	17
5 Software Programming	18
5.1 IMU	18
5.2 Distance	19
5.3 Esp32 Connection	20
5.4 Motor Control	21
5.5 PID controller	22
6 Experiment	25
6.1 Maximum Angle Experiment	25
6.2 Self Adaptive Experiment	26
7 Conclusion	29
8. Reference	30

1 INTRODUCTION

1.1 Background and Significance

The concept of human robots have been a fantasy for many people in the world. Early as in 1495, Leonardo Da Vinci invented a mechanical knight, which can sit, stand, raise its visor, and move its arm - the first humanoid robot ever documented. [1] Numerous science fantasy books and movies imagine such of a human robot. For example, Star Wars had imagined a day where humanoid robots such as C-3PO can be created. A robot that can talk, walk, run, hold, or basically acts like a human. And this is coming true. In the last few years, human robots have flourished. With an estimated market of \$1.6 billion in 2023, it is estimated to rise to more than \$2.3 billion in 5 years. [2]

Human robots is still in development. But a few have escaped prototype phase, and entered into the real world as companions, bartenders, concierges, deep sea divers, and assistants in manufacturing. [2]



Figure 1: Ameca, one of the world's best humanoid robots

For a human robot, any part of the system needs careful consideration. For robots that focuses more on movement ability, including needing the ability to move through rough terrain and great vertical distance. These robots may take the role to traverse through mountain area, or rescue people in a building. The traditional wheels may not be a good option. Human-like legs can traverse through a more complex environment. So, a good leg, or more specific, a good ankle will help the robots to traverse better in complex environments.

To learn how to build an ankle, I turned to the best teacher, nature. The human ankle is the closest, natural mechanic tuned by the nature itself. [3] Through mimicking a human ankle, I designed my own bionic ankle. For this bionic ankle, I will want to let to be able to move through complex terrain, and be able to pass through uneven ground easily.

1.2 Predecessors

1.2.1 An Energy Storage and Return Bionic Feet

An Energy Storage and Return(ESR) Bionic Feet is a bionic feet that does not use any additional energy supplies, but relies on the gait mechanics stored energy and release. [4]

The whole structure generally consists of three blades, which can deform, and store elastic energy during impact, to release during push-off phase. During the process, the foot does dorsiflexion movement with the help of the deformation of the lower and middle blade, and plantar flexion movement with the help of the defromation of the middle and upper blade, simulating those of a real human ankle.

It has the advantage of being lightweight, simple, and aids walking.

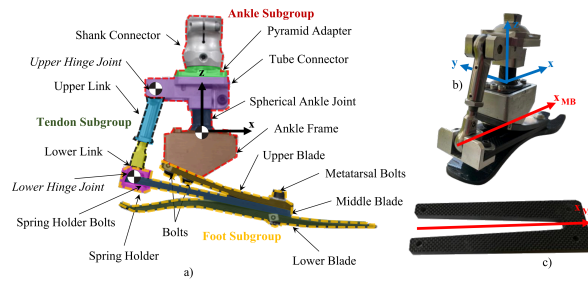


Figure 2: ESR feet diagram

ESR Bionic feet has the advantage of being lightweight, simple, and aids walking. It can be helpful on situations where heavy walking is not needed, which relying on the energy given through the gait process is enough, namely for bionic feet on some amputators. For a better propulsion by the feet, an additional thrust in the ankle is needed. This kind of design gradually evolves into the propulsive bionic feet, giving additional energy, provided by batteries, to the user.

1.2.1 Hugh Herr's bionic ankle

One predecessor of the propulsive bionic ankle is Hugh Herr's bionic ankle. [5]

The structure of the bionic ankle consists of a similar blade mechanic that stores and releases energy like the ESR feet. In addition to the blade mechanic, this model of the bionic ankle adds an additional motor to provide more energy for the ankle to move.

Hugh Herr's bionic ankle design focuses on the dorsiflexion and plantar flexion of the leg, and only focus on the assistance of walking gait. This kind of bionic ankle is lightweight, saves energy, and have the ability to navigate flat terrain.

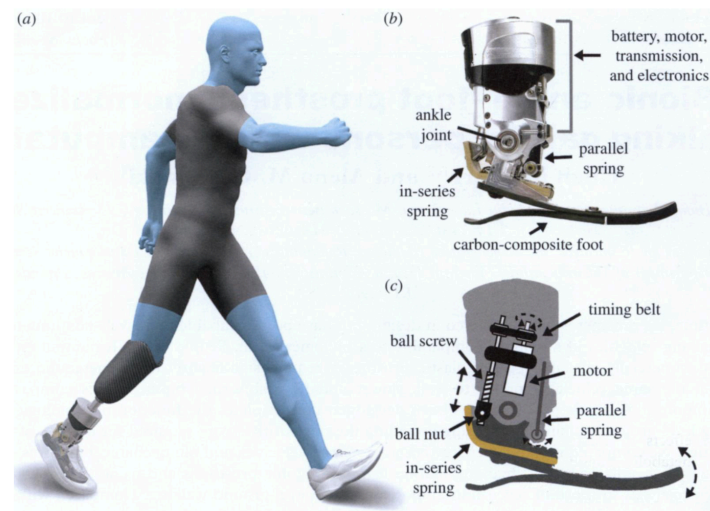


Figure 3: Hugh Herr's bionic leg diagram

However, Hugh Herr's design fails to allow the inversion and eversion movement of the ankle, and also decreasing the stability and balance of the leg. Without the inversion and eversion movement of the ankle, the foot will have some problems dealing horizontal sloping. Without inversion and eversion, the robot will have to incline the whole body, making it easier to slide of the slope.

1.2.2 A Multi-loop Ankle Foot

Another predecessor of the bionic ankle is the multi-loop Ankle Foot. [6]

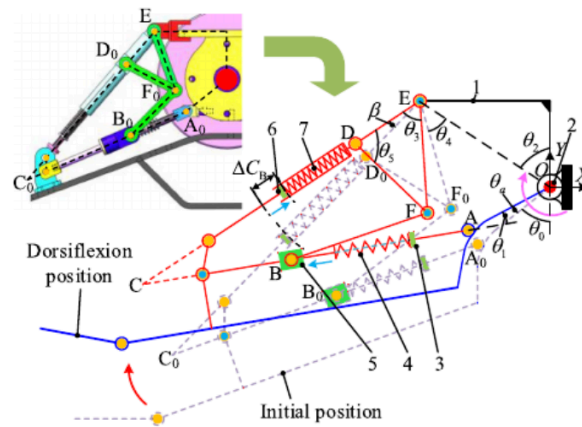


Figure 4: The Multi-loop Ankle Foot

This multi-loop ankle foot also conserves much energy and helps the amputee to move swiftly in dorsiflexion and plantar flexion, but it still lacks the ability to move in complex terrain. This kind of ankle foot have similar advantages and disadvantages as Hugh Herr's bionic ankle: both are good at energy saving and gait cycle, while both are bad at its ability to adapt to rough terrain.

1.3 Improvements

In my bionic ankle, I added

1. an additional motor to control not only the leg's dorsiflexion and plantar flexion, but also inversion and eversion movement of a leg.
2. a self adaptive program for it to aid walking using the combination of IMU and laser distance sensors.

These two factors can help the robot to balance itself better where the ground is uneven in all directions.

2 THEORETICAL DESIGN

2.1 Correspondence to a real human leg

My main goal in the design of the bionic ankle is to simulate the components of the human leg. In the final design, I simulated the bone system, joint system, muscle system, cartilage and ligaments. Those are the following corresponding parts in my bionic ankle and the real human ankle: [7]

Mechanical Ankle	Human Ankle	Function
Spring System	Articular Cartilage, Ligaments	Absorb shock, provide connection support
3D-printed Leg	Bones: Fibula, Tibia	Support the body and the drive system
Universal Joint(Block)	Bone: Talus	Provide surface to the two joints
Universal Joint(Upper Yolk)	Upper Ankle Joint	Allows dorsiflexion and plantar flexion
Universal Joint(Lower Yolk)	Lower Ankle Joint	Allows inversion and eversion
Drive System	Muscles: fibularis longus, tibialis anterior	Controls the movement of the ankle
3D-printed Foot	The whole foot	Provides a large surface for balancing the body

Table 1: Corresponding Parts of the Bionic Ankle and Human Ankle

The following diagrams compares the mechanical ankle with the human ankle. It can be seen that many elements are similar.

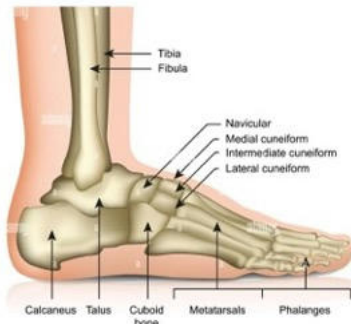


Figure 5: Side view of the Human Ankle's Bone System

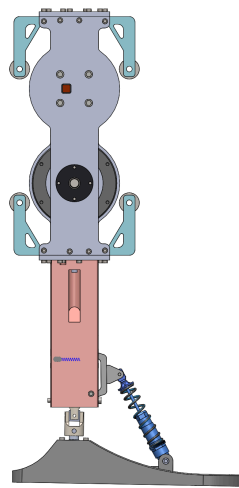


Figure 6: Side view of the Bionic Ankle

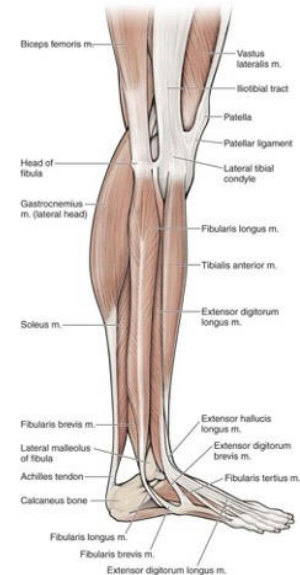


Figure 7: Side view of the Human Ankle's Muscle System

In the following sections, I will explain each structure of the bionic ankle and its purpose in correspondence to a human ankle.

2.2 Spring System

The spring system acts as a shock absorber to the system, similarly the articular cartilage in the human ankle. [8] It will allow the walking, running, and jumping process more smooth, and decrease the impact that the human / robot will experience.

It provides support to the foot, similarly to ligaments. With the spring system, the leg will be more stable. Without the spring system, the leg is only attached to the joint, reach means that the foot's motion cannot be restricted.

Additionally the spring system limits the movement of the foot. There is two springs attached to the leg, which limits the dorsiflexion, planar flexion, inversion, and eversion. This function parallels with some muscles in the human ankle.

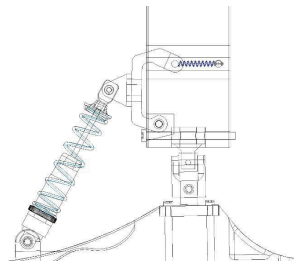


Figure 8: The demonstration of the Spring System in the ankle

2.3 3D-printed Legs

The 3D printed legs provide support to the drive system of the leg, which acts like the bones of a human body, which also provide attachment to the muscles. In my human ankle model, the 3D printed legs correlate to the bones Fibula and Tibia. Different from the human body, the 3D printed legs cannot rotate. It is an immovable object. This is due to my focus on simulating the ankle.

The lower leg length of the bionic ankle is similar to the real human leg, which is around 500 mm.

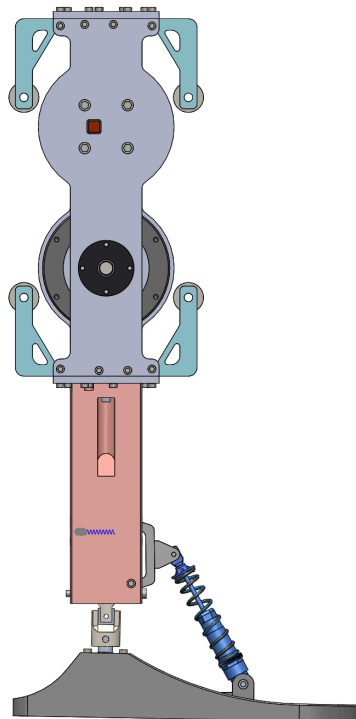


Figure 9: The demonstration of the leg in the ankle

2.4 Universal Joint

The universal joint is a mechanical component that allows two-dimensional movement, similarly to the ankle's movement. It contains three parts: upper Yolk, block, and lower Yolk. The upper Yolk and the lower Yolk allows different directions of movement. This corresponds to the ankles joint, which contains an upper ankle joint and a lower ankle joint. The upper ankle joint, formed by the Fibula, Tibia, and surface of Talus, controls the dorsiflexion and the plantar flexion. The mechanical ankle joint is formed by the 3D-printed leg and block. The lower ankle joint, formed by the talus and the bones of the foot, controls the inversion and eversion movement. The mechanical ankle joint is formed by the 3D-printed foot and block. The universal joint is like the Talus in the human body, forming the two joints that allow the leg to move.

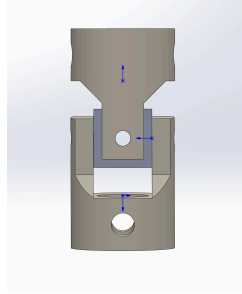


Figure 10: The demonstration of the joint in the ankle

2.5 Drive System

The drive system consists of two motors, where they simultaneously control the dorsiflexion-plantar flexion action, and the inversion-eversion action. The drive system acts like fibularis longus and tibialis anterior, which also simultaneously controls these directions.

2.6 3D-printed Foot

The 3D-printed foot is an immovable object connected to the universal joint. Although the foot is a complex structure itself, it is not the important feature that this paper hopes to consider. So, the bones, muscles, and other parts in the foot are not considered in the creation of the bionic ankle.

The foot's dimensions corresponds to the actual human foot, which has a length of 260 mm and a width of 94 mm.

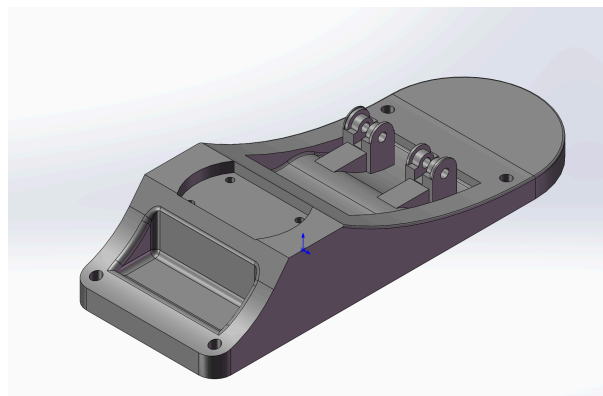


Figure 11: Foot

2.7 Conclusion

The bionic foot simulates the actual ankle very well, and simplifies the complex structures of the foot and the leg.

The joint, muscles controlling the movement of the joint, the important bones, and the connectors all can correspond to the actual foot.

3 STRUCTURAL DESIGN

3.1 Introduction

The whole ankle consists of three parts: the drive leg, the connector leg, and the foot:

1. The drive leg consists of two motors, and guide wheels.
2. The connector leg consists of tension springs, and universal joint.
3. The foot is the movable part of the whole ankle.

3.2 Drive Leg

For the bionic ankle, I will want the statistics be as close to a real human ankle.

An average human calf is around 500mm. So, for the drive leg, I estimated it to be a length of 250mm. The drive legs is the place to put two motors, so it cannot be too short in length.

I first need to design two boards that can fix the motors. To make work easy, the two boards are set to be identical. There will be two parts of the board: the motor part, where a motor is attached to this side, and the axle part, where the axle is attached to it. The first board's motor will correspond to the second board's axle. The first board's axle will correspond to the second board's motor.

The motor has a diameter of 89mm. I estimated that I need 10 mm of spacing between the motors for it to not interfere. There also need some space at the top of the board for connection to top boards and bottom boards. After calculation, the motor's center is set to be 65.5 mm away from the top of the board. To protect the motor, I also designed a circular baffle on the board to cover the motor. This circular baffle has a radius of 47.5mm, which covers the motor's 89 mm diameter. Additionally, a small hole must be at present for the motor's external DIP switch, which is used to control the ID of the motor. Moreover, on the other end of the sideboard, there must be a hole for the axle, which is set for 19mm.

The final side board is as followings:

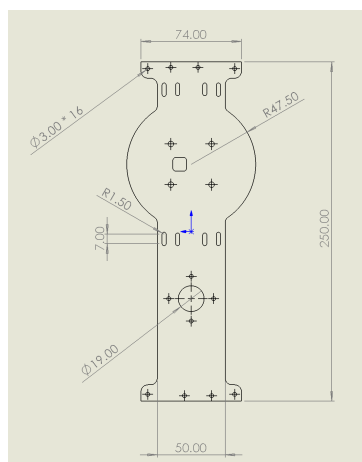


Figure 12: The Side Board of the Drive Leg

The top boards and bottom boards of the drive leg are different. Both boards need to have a support for the guide wheels, an L-shaped structure with a hole to put in the guide wheels.

The top board is used to connect with the upper leg (either another machine or human upper leg). There is no additional structure on the top of the leg. The upper leg is as follows.

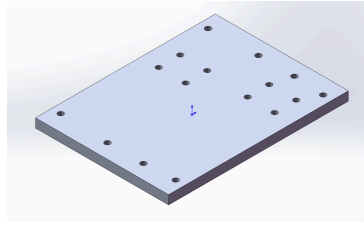


Figure 13: The Upper Board of the Drive Leg

The bottom board also need to connect with the bottom leg, making the placing of screws much harder. The bottom board is as follows.

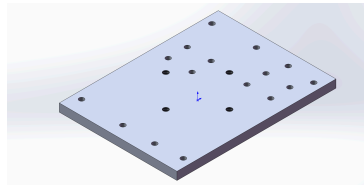


Figure 14: The Bottom Board of the Drive Leg

The guide wheels are attached to the top and bottom boards through the supports. They are used to guide the steel wire rope, which ultimately will move the leg.

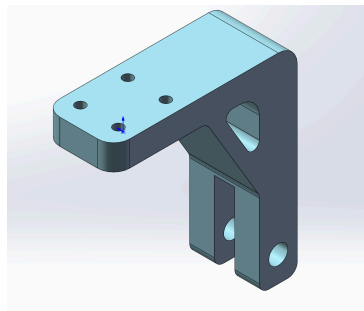


Figure 15: The Support for the Guide Wheel

Because the motors emits large vibrations, as it pulls the strings, a damping is needed to increase the load of the motor and decrease the vibration of the motor. The damping consists of an adjustable groove to change its force onto the motor, and a piece silica gel to increase the friction.

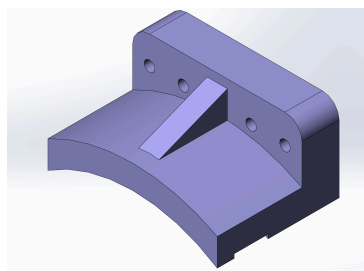


Figure 16: The Damping

The axle of the drive leg consists of a 3D printed part and a steel part. The steel part is fixated onto the board, while the 3D printed part is where the strings are located. For the steel part, it needs to be able to fix onto the board, and decrease the friction with the board.

Using a track roller, the sliding friction between the board and the axle is changed into rolling friction, effectively decreasing the friction.

Using a flange bearing, the axle is able to locate onto the board. Additionally, a cover is used to fixate the flange bearing, therefore fix all of the axle into place.

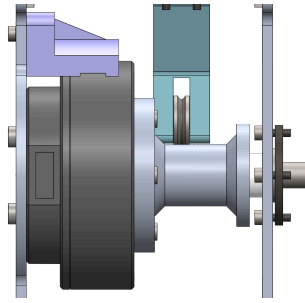


Figure 17: The Diagram of the Axle

3.3 Connector Leg

The connector leg mainly consists of a 3D-printed block, which connects the foot and the drive leg. One notable feature is the four slanted holes on its four sides, which is used to screw in the screws.

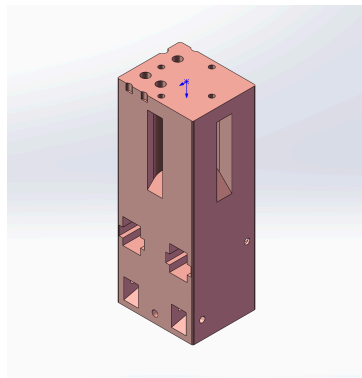


Figure 18: The connector leg

The lower part of the connector leg is where the tension springs are placed. There are two springs connected with a tension spring, used to absorb the shock from the leg. To attach the two springs and the tension springs together, there are two connectors for it. Each connector consists of a triangular connector to connect the spring, then a 3D printed connector for the tension spring.



Figure 19: The connectors

The two connector parts are fastened by screws, for which in static position, the screw head touches the 3D printed lower leg juts right.

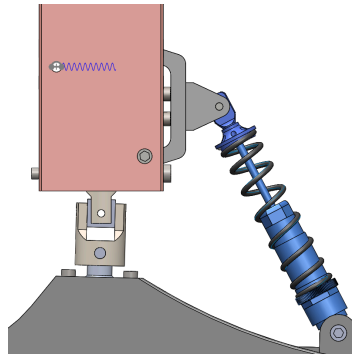


Figure 20: Demonstration of static position

Below the spring system is the universal joint, which connects the connector leg with the foot. It is a structure that allows 2D movement, and perfect for the bionic ankle.

3.4 Foot

The foot is connected to the universal joint mentioned in the lower leg part. It is attached a structure, specially designed to withstand immense forces (as it breaks in the first version of the ankle). The structure is then attached to the actual foot, which have these features:

1. It has two structures on the top of the foot to connect the springs.
2. There are four holes that for the steel wire ropes.
3. The bottom of the foot craved for a IMU.
4. The whole structure mimics a human foot

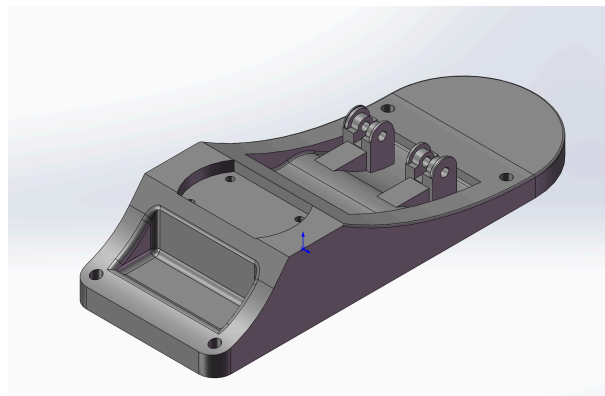


Figure 21: Foot

3.5 Wire rope

The string is the most trickiest part in the whole design of the bionic ankle. In Solidworks, it is impossible to create a good enough model to plan out how to tie the strings.

After experimenting with the tying of the strings myself, I found out that it requires a great deal of strength to loosely tie the wire rope onto the four ends of the foot.

After searching solutions to tying rope, I found two objects that can help: Wire sleeve and One way wire tighter. Wire sleeve can solve the problem of tying the wire rope to the spring. One way wire tighter can easily fix the steel rope to the four corners of the foot.



Figure 22: One way wire tighter



Figure 23: wire sleeve

Additionally, I added a coating for the most of the segments to increase friction, leaving 1 segment not coated to be able to adjust the tightness of the rope.

3.6 Conclusion

The final design of the ankle is the following diagram:

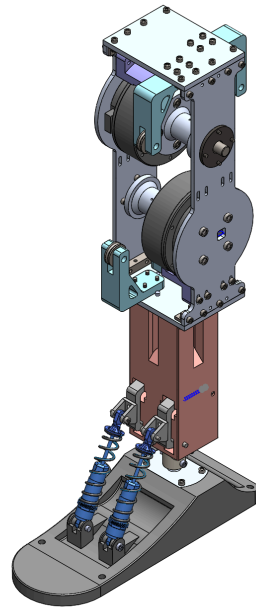


Figure 24: Bionic Ankle Diagram

In my bionic ankle, the following parts are used:

Part	Type	Numbers
MS9015	bought	2
TTL - RS485	bought	1
steel axle	cut	2
wider axle	3D printed	2
damping	3D printed	2
track rollers	bought	2
flange bearings	bought	2
flange bearings cover	laser cutting	2
guide wheels	bought	4
guide wheels supporter	3D printed	4
side board	customized	2
top board	customized	2
bottom board	customized	2
supporting leg	3D printed	1
springs	bought	2
tension springs (different statistics)	bought	6
tension spring connector	customized	2
spring connector	bought	2
universal joint	bought	1
universal joint connector	customized	1
foot	3D printed	1
JY62	bought	1
VL53L0X	bought	4
TCA9548A	bought	1
esp32 board	bought	3

Table 2: All of the parts used in the bionic ankle

4 HARDWARE DESIGN

4.1 Introduction

For the hardware of the ankle, I used

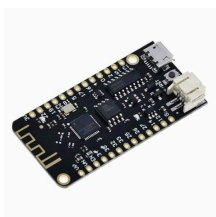
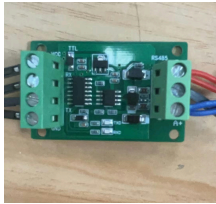
Name	Type	Image	Usage
JY62	IMU		Give the pitch, roll, and yaw of the foot
VL53L0X	ToF sensor		Give the distance to the nearest object
TCA9548A	switch		Transmits multiple devices' signal
esp32	control board		Controls other devices
MS9015	motor		Turns the axle
24V battery	battery		Provides voltage
TTL-RS485	communication conversion module		Converts different communication protocols

Table 3: Corresponding Parts of the Bionic Ankle and Human Ankle

4.2 JY62

JY62 is a IMU that can provide the pitch, roll, and yaw of the foot through fusion algorithm. The stable output of the IMU can be used to do PID controller.

4.7 VL53L0X

VL53L0X is a laser ToF sensor that detects the distance from itself to an object in front of it. It has the ability to detect up to 2 meters.

Since I intend to have 4 VL53L0X, a I2C switch is necessary to send data to the esp32 board. TCA9548A is a I2C switch that can connect different devices and send the directed channel to the main board.

4.5 Esp32

The purpose of esp32 is for its wireless connection to other esp32. For the IMU and the ToF distance sensor, I intended to attach to the bottom of the foot. Using a wireless connection can avoid complex wiring.

4.2 Motor MS9015

A good motor is a motor that has a large enough torque to pull the ankle. The human ankle has a torque around 100 - 200 Nm . Let the radius of the axle be r , the distance from the joint to the front string be d_1 , the distance from the joint to the back string be d_2 , and the torque of the motor be T_{motor} .

The force of the string can be calculated as follows:

$$F_{string} = \frac{T_{motor}}{r} \quad (1)$$

The force of the string is estimated to act perpendicularly on the foot, so the torque of the ankle can be calculated as follows:

$$T_{ankle} = F_{string} \times d_1 + F_{string} \times d_2 = F_{string} \times (d_1 + d_2) = T_{motor} \times \frac{d_1 + d_2}{r} \quad (2)$$

The motor MS9015 has the following statistics:

1. a rated torque of 1.55 N m
2. a rated voltage of 24 V
3. a rated speed of 100 rpm
4. a max power of 17 W

Also, the radius of the motor is 0.012 m; the distance from the front string to the joint is 0.05951 m; the distance from the back string to the joint is 0.12256 m. Inputting the numbers, the torque of the ankle is 23.51 N m.

The torque and power of MS9015 is strong enough to give enough force to move the axle. A smaller one cannot easily move the foot. A larger one makes the whole ankle too heavy.

Other hardware are needed to use the motor MS9015.

A 24V battery can provide enough voltage to power the motor. The motor uses RS485 protocol. I used esp32 board to control the motor, so a TTL – RS485 translator is needed to communicate with the motor.

5 SOFTWARE PROGRAMMING

5.1 IMU

For my first version of the bionic ankle, I choose MPU6050 as the IMU implanted in the foot.

MPU6050 will give 6 variables:

1. x direction acceleration (a_X)
2. y direction acceleration (a_Y)
3. z direction acceleration (a_Z)
4. x direction angular velocity (g_X)
5. y direction angular velocity (g_Y)
6. z direction angular velocity (g_Z)

Using only accelerometer, it is possible to calculate the pitch, and roll according to the following equations:

$$r = \arctan\left(\frac{a_Y}{\sqrt{a_X^2 + a_Z^2}}\right) \quad (3)$$

$$p = \arctan\left(\frac{a_X}{\sqrt{a_Y^2 + a_Z^2}}\right) \quad (4)$$

However, the roll and pitch are only correct in a ideal situation. Accelerometer measures all forces, including gravity, making it inaccurate. The IMU is the more accurate than the accelerometer, but IMU requires integration of data, and easily can cause a gyro-drift where the error of the IMU sums up to cause a great problem.

A solution to get accurate pitch and roll is to fuse the accelerometer's data and the I Mu's data in the following equation:

$$r_f = w \times (r_i + g_X) + (1 - w) \cdot r_a \quad (5)$$

$$p_f = w \times (p_i + g_X) + (1 - w) \cdot p_a \quad (6)$$

This combination can accurately calculate roll and pitch, but I found a better IMU JY62 that can give pitch, roll, and yaw directly. So, I changed into JY62 instead of MPU6050.

```
if (Re_buf[0] == 0x55)
{
    switch (Re_buf[1]) {
        case 0x51:
            a[0] = (short(Re_buf[3] << 8 | Re_buf[2])) / 32768.0 * 16;
            a[1] = (short(Re_buf[5] << 8 | Re_buf[4])) / 32768.0 * 16;
            a[2] = (short(Re_buf[7] << 8 | Re_buf[6])) / 32768.0 * 16;
            T = (short(Re_buf[9] << 8 | Re_buf[8])) / 340.0 * 36.25;
            senddata.ax = 9.8*a[0];
            senddata.ay = 9.8*a[1];
            senddata.az = 9.8*a[2];
            datacount++;
            break;
        case 0x52:
            w[0] = (short(Re_buf[3] << 8 | Re_buf[2])) / 32768.0 * 2000;
            w[1] = (short(Re_buf[5] << 8 | Re_buf[4])) / 32768.0 * 2000;
            w[2] = (short(Re_buf[7] << 8 | Re_buf[6])) / 32768.0 * 2000;
            T = (short(Re_buf[9] << 8 | Re_buf[8])) / 340.0 * 36.25;
            senddata.wx = w[0];
            senddata.wy = w[1];
            senddata.wz = w[2];
            datacount++;
            break;
        case 0x53:
            angle[0] = (short(Re_buf[3] << 8 | Re_buf[2])) / 32768.0 * 180;
            angle[1] = (short(Re_buf[5] << 8 | Re_buf[4])) / 32768.0 * 180;
            angle[2] = (short(Re_buf[7] << 8 | Re_buf[6])) / 32768.0 * 180;
            T = (short(Re_buf[9] << 8 | Re_buf[8])) / 340.0 * 36.25;
            senddata.pitch = angle[0];
            senddata.roll = angle[1];
            senddata.yaw = angle[2];
            datacount++;
            break;
    }
}
```

Figure 25: The communication process of JY62

5.2 Distance

Using the 4 distance sensors attached to the four corners, I can get the information about the pitch and roll of the foot relative to the ground.

Receiving the data from the 4 identical VL53L0X requires a I2C switch TCA9548A. Using this I2C, it is easy to receive the data from the 4 VL53L0X by changing the channel of the bus.

```
void PCA9548A(uint8_t bus) {  
  Wire.beginTransmission(0x70);  
  Wire.write(1 << bus);  
  Wire.endTransmission();  
}
```

Figure 26: The function to change the channel of the switch

After receiving the distance from the ground, the next step is to combine the data to calculate the angle.

To calculate the angle of between the foot and the ground, consider the 2D demonstration of the foot and the ground.

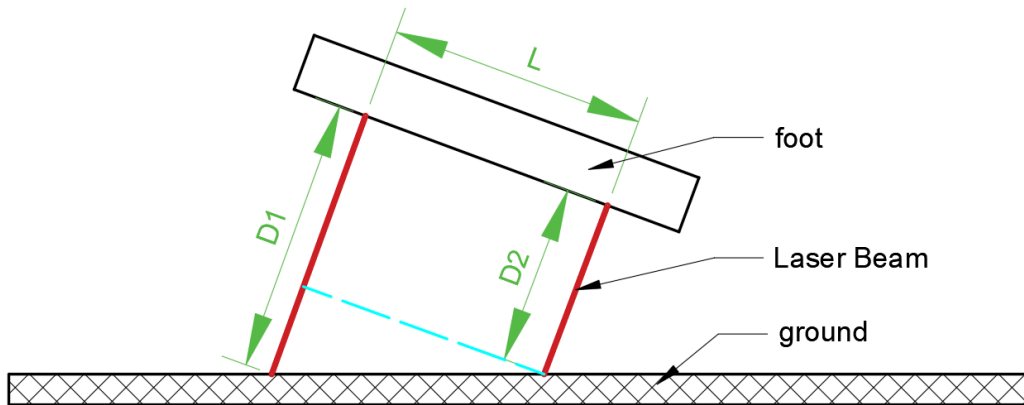


Figure 27: 2D demonstration of the foot and the ground

The angle between the foot and the ground can be calculated using arctan function. Note that Δd can be negative to demonstrate the direction the foot is slanting from the ground.

$$m\angle ABC = \arctan\left(\frac{\Delta d}{l}\right) \quad (7)$$

```
double Find_angle(double d1, double d2, int length) {
    double side = d1 - d2;
    double a = side / length;
    // Serial.println(a);
    return atan(a) / pi * 180;
}
```

Figure 28: The code for the above equation

However, the distance sensor is not consistent enough to use it along in calculating the angle between the foot and the ground, so the average of two distance sensors are used together to provide a more consistent data for the distance.

```
two Find_PR(int LU, int LD, int RU, int RD) {
    double h1 = (LU + RU) / 2;
    double h2 = (LD + RD) / 2;

    double v1 = (LU + LD) / 2;
    double v2 = (RU + RD) / 2;

    two ans = {Find_angle(h1, h2, length), Find_angle(v1, v2, width)};

    return ans;
}
```

Figure 29: The overall calculation for pitch and roll

The result of the pitch and roll calculation is as follows:

It is obvious that the pitch and roll is too instable. A filtering mechanics is needed to stabilize the pitch and roll received. I used the average of the last n data to filter out the extreme cases.

```
if (num_data < points) {
    for (int i = 0; i < 4; i++) {
        window[i][num_data] = d[i];
    }
    num_data++;
}
else {
    int average[4];
    for (int i = 0; i < 4; i++) {
        average[i] = window[i][0];
    }
    for (int i = 0; i < 4; i++) {
        for (int j = 1; j < points; j++) {
            average[i] += window[i][j];
            window[i][j - 1] = window[i][j];
        }
    }

    for (int i = 0; i < 4; i++) {
        window[i][points - 1] = d[i];
        average[i] /= points;
    }

    two angles_2 = Find_PR(average[1], average[3], average[0], average[2]);
}
```

Figure 30: The process of filtering

After filtering the reaction speed of the pitch and roll will decrease, but the stability of the pitch and roll will increase drastically.

After calculating the pitch and roll of the foot relatively to the ground, the data needs to send to the PID controller using an Esp32 WiFi connection.

5.3 Esp32 Connection

The reason I choose esp32 as the main controller is because it has a built-in wireless connection for its board. An esp32 (master) can send data to another esp32 (slave) with its given MAD address.

I need to send the pitch and roll of the foot and the relative pitch and roll of the foot to the ground. To send the package of the data, I created a structure called data_struct to contain the id, pitch, roll, and others that the IMU can calculate but the distance sensors cannot calculate. The id is used to detect which esp32 board send the data.

```
typedef struct data_struct {
    int id;
    float pitch;
    float roll;
    float yaw;
    float ax;
    float ay;
    float az;
    float wx;
    float wy;
    float wz;
} data_struct;

data_struct board0;
data_struct board1;

data_struct board_struct[2] = {board0, board1};

// Create a data_struct called myData
data_struct myData ;
```

Figure 31: The dataholder of the receiver side

For the receiver side, aside from the container myData to receive the data, there is also an array to hold the data sent by different id.

After setting up the connection between esp32 boards, the transferring of the data only requires some functions predefined by esp32.

5.4 Motor Control

The motor MS9015 built in many different instructions, such as turning the motor to a specific angle, stopping the motor, reading the angle of the motor, incrementing the motor... I choose to use the instruction “incrementing the angle” as the control function for the motor.

The motor requires a 14 byte data to tell how it should to move. For instruction 16, incrementing the position of the motor, the 14 byte of data are instructed as follows:

Header (5 byte)		
CMD[0]	header	0x3E
CMD[1]	instruction	0xA8
CMD[2]	id	0x01 - 0x20
CMD[3]	data length	0x08
CMD_SUM	check	sum of previous bytes

Table 4: Header of the whole data

Data (5 byte)		
DATA[0]	increment low byte	$*(\text{uint8_t})(&\text{angleIncrement})$
DATA[1]	increment byte	$*((\text{uint8_t})(&\text{angleIncrement}) + 1)$
DATA[2]	increment byte	$*((\text{uint8_t})(&\text{angleIncrement}) + 2)$
DATA[3]	increment high byte	$*((\text{uint8_t})(&\text{angleIncrement}) + 3)$
DATA_SUM	check	sum of previous bytes

Table 5: Header of the whole data

The variable angleIncrement represents 0.01° angle increment / LSB, using $-36000 - 36000$ to represent $-360^\circ - 360^\circ$. It has a sign to determine the direction the motor will turn. The angleIncrement is separated into four parts, DATA[0] - DATA[3], to be sent into the motor.

```
void angle_increas_send(int ID,int32_t i_angle)
{
    senddate[0]=0x3E;
    senddate[1]=0xA7;
    senddate[2]=ID;//电机ID
    senddate[3]=0x04;
    senddate[4]=senddate[0]+senddate[1]+senddate[2]+senddate[3];

    senddate[5] = (uint8_t)(i_angle & 0xFF);//数据低八位
    senddate[6] = (uint8_t)((i_angle >> 8) & 0xFF);//数据高八位
    senddate[7] = (uint8_t)((i_angle >> 16) & 0xFF);//数据高八位
    senddate[8] = (uint8_t)((i_angle >> 24) & 0xFF);//数据高八位
    senddate[9]=senddate[5]+senddate[6]+senddate[7]+senddate[8];
    Serial2.write(senddate,10);
}
```

Figure 32: The function of motor angle increment

This function inputs the angle of the motor, and the id of the motor, to turn the motor with the specific id by the specific angle.

5.5 PID controller

The PID controller is the main algorithm to control the whole foot.

The PID controller is functions as follows:

First, the PID controller receives the error of between the desired value and the actual value. It does 3 actions to quickly and accurately decrease the error.

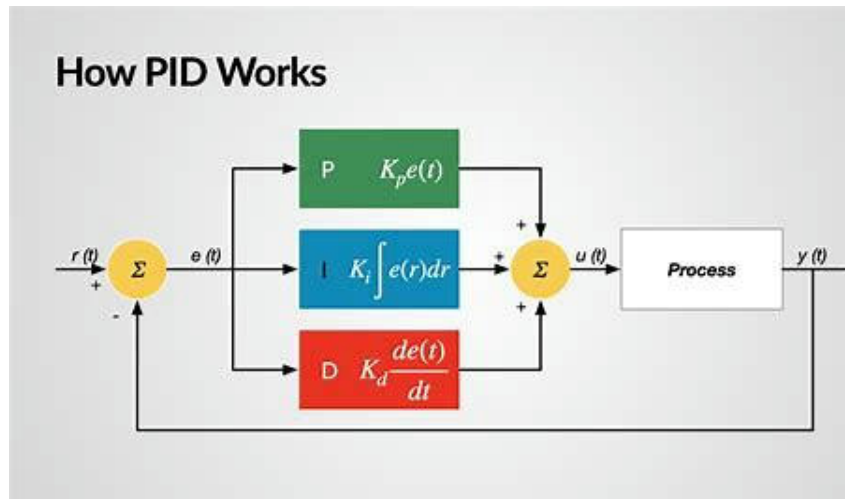


Figure 33: A diagram of the PID controller

1. Proportional: The PID controller decreases the present error proportionally. This action is quick, but cannot diminish steady-state errors.
2. Integral: The integral looks back upon previous errors to correct the steady-state errors that proportional cannot adjust.
3. Derivative: The derivative looks upon the trend of the error, and dampens the speed of increasing or decreasing to prevent overshooting and undershooting.

The equation for the PID controller using discrete time is

$$u(t) = k_p \times e(t) + k_i \times \Sigma e(t) + k_d \times (e(t) - e(t - 1)) \quad (8)$$

The three actions are combined together and given constants to control how aggressive each action is. A good constant will allow the PID controller to react quickly and accurately.

The PID controller for two motors are more complex, because there are two data, pitch and roll, to control using two motors. I assumed that the two motors' increment provides identical pull regardless of direction. This means that I need to do two PID controller for pitch and roll.

```

float PID_control (float now_date,float set_date )
{
    float err = set_date - now_date;
    err_all = err + err_all;

    if (err_all > 1000) err_all = 1000;
    if (err_all < -1000) err_all = -1000;
    float P_OUT = kp * err;
    float I_OUT = ki * err_all;
    float D_OUT = kd * (err - last_err);
    last_err = err;
    float all_out = P_OUT + I_OUT + D_OUT;
    if (all_out > 1200) all_out = 1200;
    if (all_out < -1200) all_out = -1200;
    return all_out;
}

float PID_control2 (float now_date2,float set_date2)
{
    ...
}

void PID_total (float roll, float pitch, float set_pitch, float set_roll) {
    roll_PID_angle = PID_control(roll, set_roll);
    pitch_PID_angle = PID_control2(pitch, set_pitch);
    angle_increase_send(1,-roll_PID_angle + pitch_PID_angle);
    delay(10);
    angle_increase_send(2,+roll_PID_angle + pitch_PID_angle);
    delay(10);
}

```

Figure 34: The code of PID controller

Note: PID_control and PID_control2 are 2 identical function using different sets of variables.

This is the following result of the PID controller. The reaction time of the PID controller is very fast.

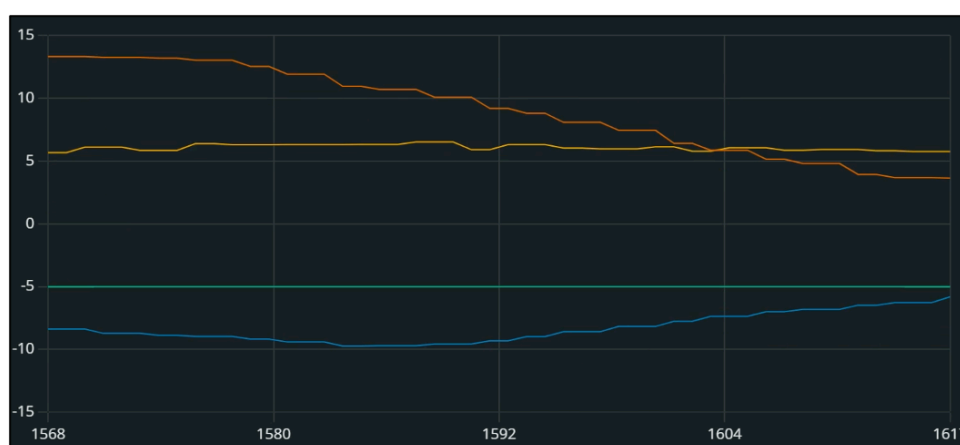


Figure 35: The result of PID controller

6 EXPERIMENT

6.1 Maximum Angle Experiment

In this experiment, I am going to measure the maximum angle each direction the foot can move. The purpose of the experiment is to show the flexibility of the ankle, that it can achieve the common movements that humans can do. The limitations of the maximum angle of the leg are caused by the four strings length and the parallel springs length.

The procedure is as follows:

1. I use my hands to turn the motor such that the foot is in the extreme position. Even if the foot can still move further, it is unnecessary because the strings are too strained to be stable and safe.
2. I take a picture of the lower part of the foot with the bottom face of the supporting leg perpendicular to the photo frame, and the bottom face of the foot perpendicular to the photo frame.
3. I use Tracker to analyze the photo:
 1. Take the bottom of the supporting leg as the x-axis, since it is perpendicular to the photo frame, the bottom of the supporting leg is represented by a line.
 2. Use the bottom of the foot as the other ray that locks the angle with the x-axis. The intersection between the line formed by the bottom of the foot and x-axis is the center of the protractor.

These are the pictures I used to measure the angle:

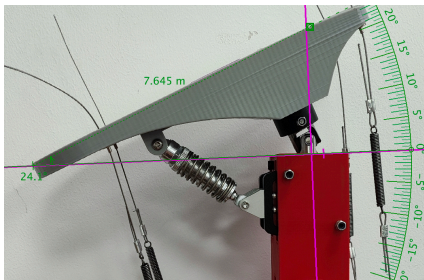


Figure 36: Dorsiflexion

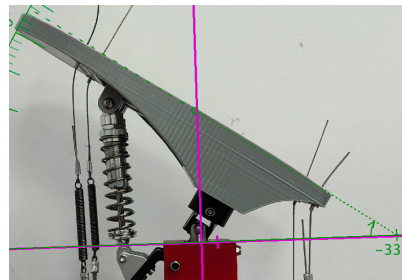


Figure 37: Plantar Flexion

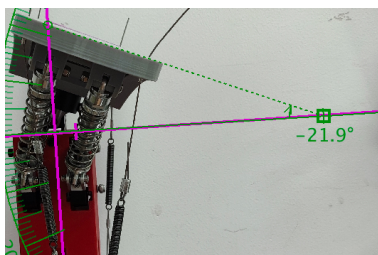


Figure 38: Inversion

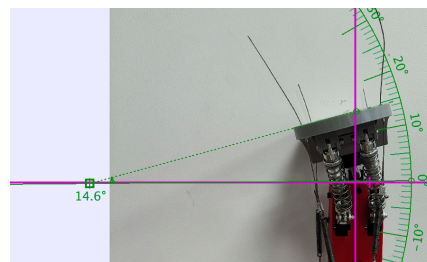


Figure 39: Eversion

The average human's maximum angle in plantar flexion, dorsiflexion, inversion, and eversion are 40°, 20°, 23°, 12° respectively. Plantar flexion maximum angle can range from 40° - 50°; dorsiflexion maximum angle can range from 10° - 20°. I will take the closest number for my bionic ankle. [7], [9]
Using the equation

$$PE = \frac{|actual - expected|}{expected} \times 100\% \quad (9)$$

The percent error of the bionic ankle in four directions are listed below.

Movement	Maximum Angle	Human's Maximum Angle	Percent Error
Plantar Flexion	33.7°	40°	16%
Dorsiflexion	24.1°	20°	20.5%
Inversion	21.9°	23°	4.7%
Eversion	14.6°	12°	22%

Table 6: All of the parts used in the bionic ankle

With the comparison to the average human data, the bionic ankle's error is quite low, mostly being around 10% - 20%. The bionic ankle have similar flexibility compared with the actual human ankle. This means that the ankle can achieve different gaits and methods to move like a human.

6.2 Self Adaptive Experiment

The ability to adapt to multiple angles of the ground is an important feature of the bionic ankle. Using VL53L0X, the inclination of the ground can be known. This experiment aims to show the bionic ankle's ability to follow the inclination of the ground.

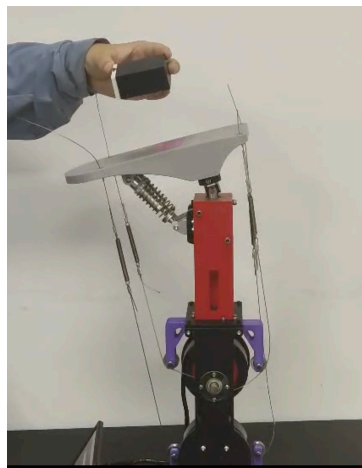


Figure 40: The process of turning

Using a block to act as the ground, I turned in all 4 directions to test whether the foot can follow the ground. It is noted that there is a limit for the angle to turn in order to protect the strings. In the below figure, the yellow and green lines represent the ground's pitch and roll, while the red and blue lines represent the foot's pitch and roll.

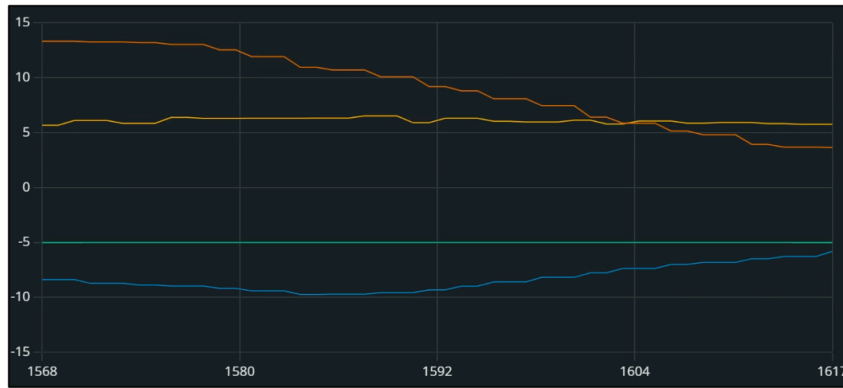


Figure 41: The PID display of the turning

The unit of the x-axis is in millisecond, showing that the foot can move quickly to reach the desired value in around 25 milliseconds. This speed can be improved further more through an more optimized PID controller algorithm and a faster motor.

Overall, this experiment shows that the foot have the ability to adapt itself to different inclination of the ground, making it able to move in complex terrains.

In this experiment, I have not yet implemented VL53L0X onto it. The next step of the self-adaptive experiment is to change the foot's model, and add VL53L0X to the four corners of the foot.

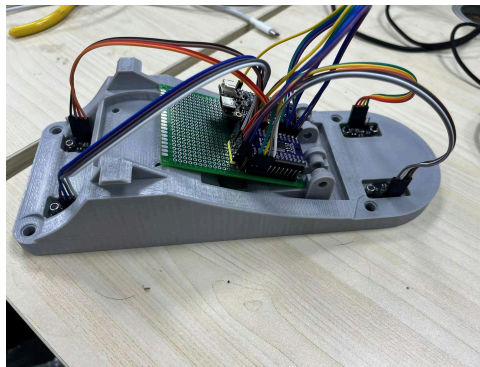


Figure 42: The new version of the foot which include VL53L0X

After adding 4 VL53L0X to the corners of the foot, the foot can follow the inclination of any ground swiftly.

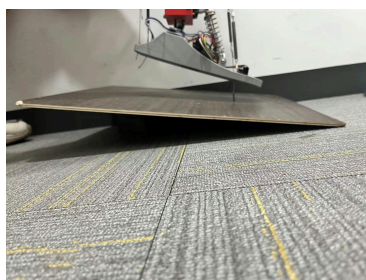


Figure 43: demo1

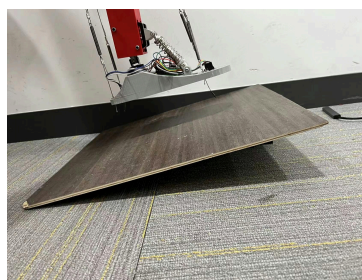


Figure 44: demo2

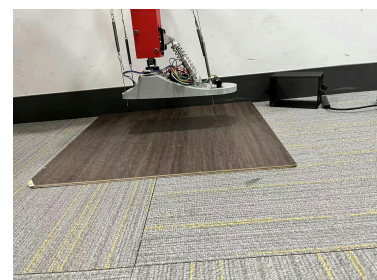


Figure 45: demo3

In any different inclination of the ground, the foot can land parallelly on the plane. This adaptation will help the foot to walk in complex environment. There can also be some additional improvements on the bottom of the foot, like adding texture to increase the grip of the ground.

Through this experiment, the foot have the ability to adapt to the different inclination of the ground.

7 CONCLUSION

My bionic ankle focuses on having 2 degrees of freedom, and the ability to self adapt to the ground below using sensors to correct the position of the foot. It provides new insights on how to interact with the ground with a foot that moves 2 dimensionally.

The bionic ankle still have some weakness.

1. The bionic ankle is still too heavy. To achieve the ability of having 2 degrees of freedom, two motors must be used in the bionic ankle. This increases the weight of the bionic ankle, while not providing enough torque to match the actual human ankle.
2. The bionic ankle has not fully incorporate the function to connect to other parts of the machine, meaning that it still needs more work to incorporate the bionic ankle into the system.
3. The bionic ankle's foot have no flexibility, which can prove to be a problem when the action of gripping is needed.

8. REFERENCE

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